

# CatevCNN1D v8 — Model Results Report

F.T · Physiological Instability Detection Engine

ICU Patient Deterioration Prediction · 3-Class Severity Output

## Dataset & Training Configuration

| Parameter           | Value                                                 |
|---------------------|-------------------------------------------------------|
| Total rows loaded   | 620,257                                               |
| Total patients      | 103                                                   |
| Features            | 41                                                    |
| Window size         | 45 rows (90 s)                                        |
| Stride              | 30 rows (60 s)   Overlap: 33%                         |
| Jitter augmentation | ±7 rows (train only)                                  |
| Batch size          | 256                                                   |
| Max epochs          | 60 (early stopped @ epoch 11)                         |
| Initial LR          | 0.0003 → reduced via ReduceLROnPlateau                |
| Class weights       | Normal: 0.7672   Critical: 1.6563   Emergency: 0.9151 |

Target Distribution (future\_label)

| Class | Label     | Count   | Share |
|-------|-----------|---------|-------|
| 0     | Normal    | 244,449 | 39.4% |
| 1     | Critical  | 118,468 | 19.1% |
| 2     | Emergency | 257,340 | 41.5% |

Patient-Level Split

| Split | Patients | Normal % | Critical % | Emergency % | Windows |
|-------|----------|----------|------------|-------------|---------|
| Train | 80       | 43.3%    | 20.2%      | 36.5%       | 30,230  |
| Val   | 10       | 21.0%    | 21.6%      | 57.4%       | 2,228   |
| Test  | 13       | 31.9%    | 10.4%      | 57.7%       | 2,664   |

## Training Progress

| Epoch | Train Loss | Train Acc | Val Loss | Val Acc | LR   |
|-------|------------|-----------|----------|---------|------|
| 1     | 1.1474     | 44.49%    | 1.0009   | 54.04%  | 3e-4 |

|    |        |        |        |        |          |
|----|--------|--------|--------|--------|----------|
| 2  | 1.0195 | 53.92% | 1.0545 | 50.27% | 3e-4     |
| 3  | 0.9658 | 58.30% | 1.1117 | 46.63% | 3e-4     |
| 4  | 0.9124 | 61.87% | 1.2188 | 42.64% | 3e-4     |
| 5  | 0.8649 | 64.62% | 1.2407 | 44.61% | 3e-4     |
| 6  | 0.8238 | 66.49% | 1.2947 | 46.41% | 1.2e-4 ↓ |
| 7  | 0.7626 | 70.52% | 1.3766 | 46.05% | 1.2e-4   |
| 8  | 0.7252 | 72.41% | 1.4319 | 43.90% | 1.2e-4   |
| 9  | 0.6972 | 73.75% | 1.4851 | 43.99% | 1.2e-4   |
| 10 | 0.6738 | 74.74% | 1.5691 | 42.64% | 1.2e-4   |
| 11 | 0.6501 | 76.04% | 1.6560 | 41.83% | 4.8e-5 ↓ |

★ Best epoch: Epoch 1 (lowest val\_loss = 1.0009) — model weights restored

### Validation Results (10 patients · 2,228 windows)

|               |                   |              |
|---------------|-------------------|--------------|
| <b>0.4671</b> | <b>0.4608</b>     | <b>54.0%</b> |
| Macro F1      | Balanced Accuracy | Accuracy     |

| Class         | Precision | Recall | F1-Score | Support |
|---------------|-----------|--------|----------|---------|
| Normal (0)    | 0.52      | 0.44   | 0.48     | 465     |
| Critical (1)  | 0.26      | 0.26   | 0.26     | 490     |
| Emergency (2) | 0.65      | 0.69   | 0.67     | 1273    |
| Macro avg     | 0.48      | 0.46   | 0.47     | 2228    |
| Weighted avg  | 0.54      | 0.54   | 0.54     | 2228    |

Confusion Matrix — Normalised (row %)

|                | Pred-Normal | Pred-Critical | Pred-Emergency |
|----------------|-------------|---------------|----------------|
| True-Normal    | 44.1%       | 13.1%         | 42.8%          |
| True-Critical  | 17.4%       | 25.5%         | 57.1%          |
| True-Emergency | 7.9%        | 23.4%         | 68.7%          |

### Test Results (13 patients · 2,664 windows)

|               |                   |              |
|---------------|-------------------|--------------|
| <b>0.6053</b> | <b>0.6144</b>     | <b>72.0%</b> |
| Macro F1      | Balanced Accuracy | Accuracy     |

| Class         | Precision | Recall | F1-Score | Support |
|---------------|-----------|--------|----------|---------|
| Normal (0)    | 0.83      | 0.86   | 0.84     | 852     |
| Critical (1)  | 0.16      | 0.27   | 0.20     | 281     |
| Emergency (2) | 0.84      | 0.72   | 0.77     | 1531    |
| Macro avg     | 0.61      | 0.61   | 0.61     | 2664    |
| Weighted avg  | 0.76      | 0.72   | 0.73     | 2664    |

Confusion Matrix — Normalised (row %)

|                | Pred-Normal | Pred-Critical | Pred-Emergency |
|----------------|-------------|---------------|----------------|
| True-Normal    | 85.9%       | 5.5%          | 8.6%           |
| True-Critical  | 23.5%       | 26.7%         | 49.8%          |
| True-Emergency | 5.6%        | 22.7%         | 71.7%          |

## Key Observations

**Normal class (test):** Strong performance — precision 0.83, recall 0.86, F1 0.84. The model cleanly separates stable patients.

**Emergency class (test):** Good performance — precision 0.84, F1 0.77. 71.7% of true emergencies correctly flagged.

**Critical class:** Hardest to classify across both val and test (F1 ~0.20–0.26). Reflects the transitional, physiologically ambiguous nature of the Critical state.

**Val vs Test gap:** Test macro-F1 (0.61) exceeds val macro-F1 (0.47). Val set has an atypical class distribution (57% Emergency) vs. test (58% Emergency, 32% Normal), contributing to the apparent gap.

**Early stopping:** Training halted at epoch 11; best weights restored from epoch 1. Indicates aggressive overfitting from epoch 2 onward — a candidate for stronger regularisation or architecture revision in v9.

## Saved Model Artifacts

| File                           | Description                                |
|--------------------------------|--------------------------------------------|
| catev_cnn_v8_model.keras       | Trained model weights                      |
| catev_cnn_v8_meta.pkl          | StandardScaler + metadata                  |
| catev_cnn_v8_report.txt        | Full evaluation report                     |
| catev_cnn_v8_history.png       | Training & validation loss/accuracy curves |
| catev_cnn_v8_validation_cm.png | Validation confusion matrix plot           |
| catev_cnn_v8_test_cm.png       | Test confusion matrix plot                 |

